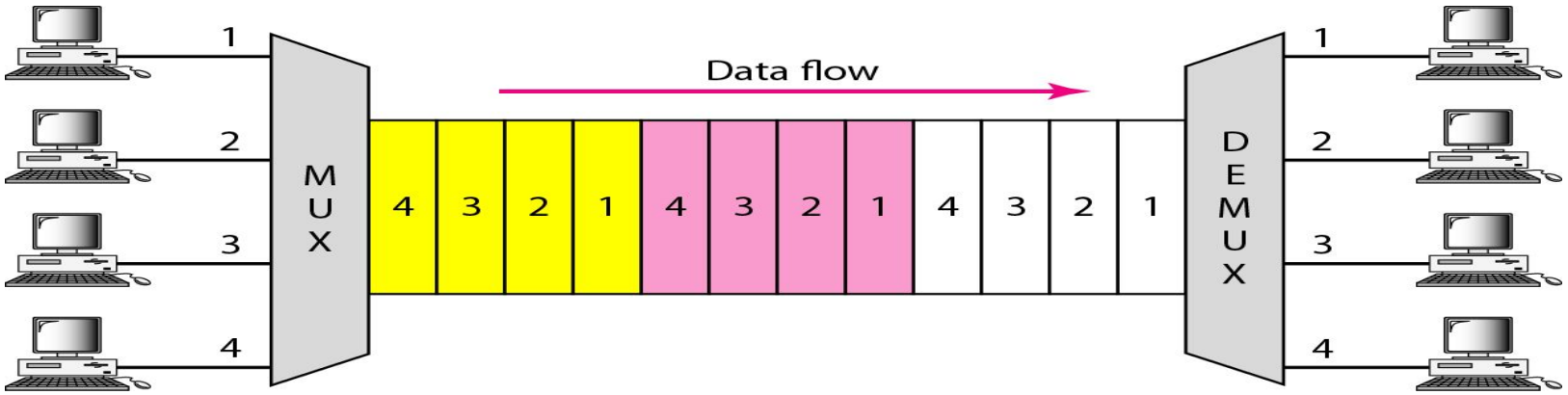
Data Communication
CSE 2211

Chapter 6 : Multiplexing-TDM

Time-division multiplexing

- *Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link.*
- *Instead of sharing a portion of the bandwidth as in FDM, time is shared.*
- *Each connection occupies a portion of time in the link.*
- *In the figure, portions of signals 1, 2, 3, and 4 occupy the link sequentially.*
- *We can divide TDM into two different schemes: synchronous and statistical.*

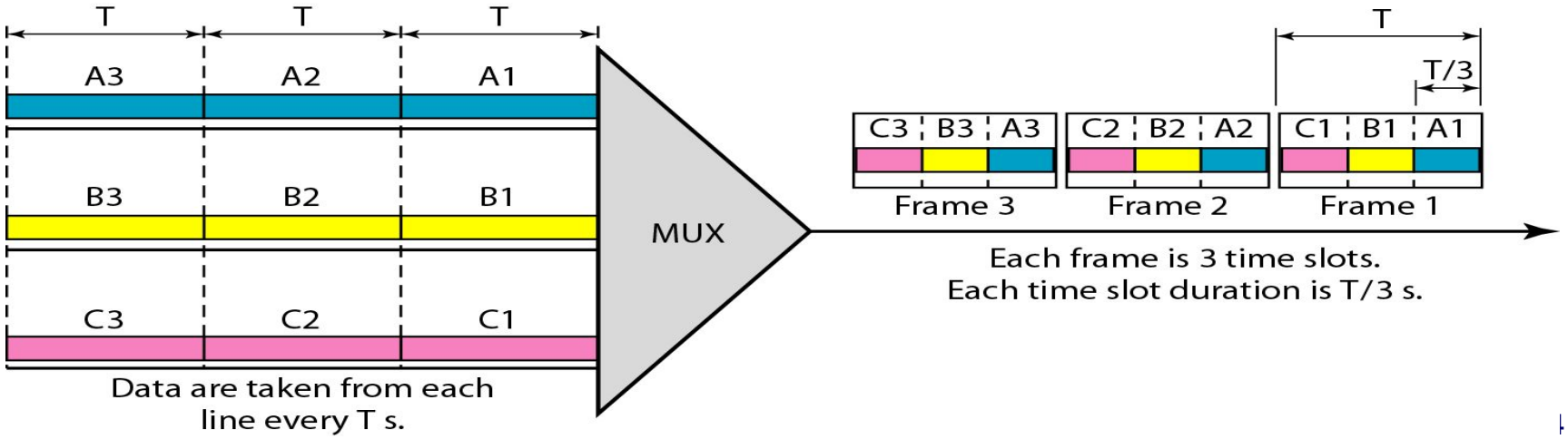


Note

We also need to remember that TDM is, in principle, a digital multiplexing technique.
Digital data from different sources are combined into one timeshared link.

Synchronous TDM

- In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.
- The data flow of each input connection is divided into units, and the link combines one unit of each connection to make a frame.
- The size of the unit can be 1 bit, or several bits.
- For n input connections, a frame is organized into a minimum of n time slots, each slot carrying one unit from each connection.



Note

In synchronous TDM, the data rate of the link is *n* times faster, and the unit duration is *n* times shorter.

Example 6.5

In Figure 6.13, the data rate for each one of the 3 input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

We can answer the questions as follows:

a. The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).

Example 6.5 (continued)

b. *The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.*

c. *Each frame carries three output time slots. So the duration of a frame is 3 × 1/3 ms, or 1 ms.*

Note*: The duration of a frame is the same as the duration of an input unit.*

Example 6.6

Figure 6.14 shows synchronous TDM with four 1Mbps data stream inputs and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

a. The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 µs.

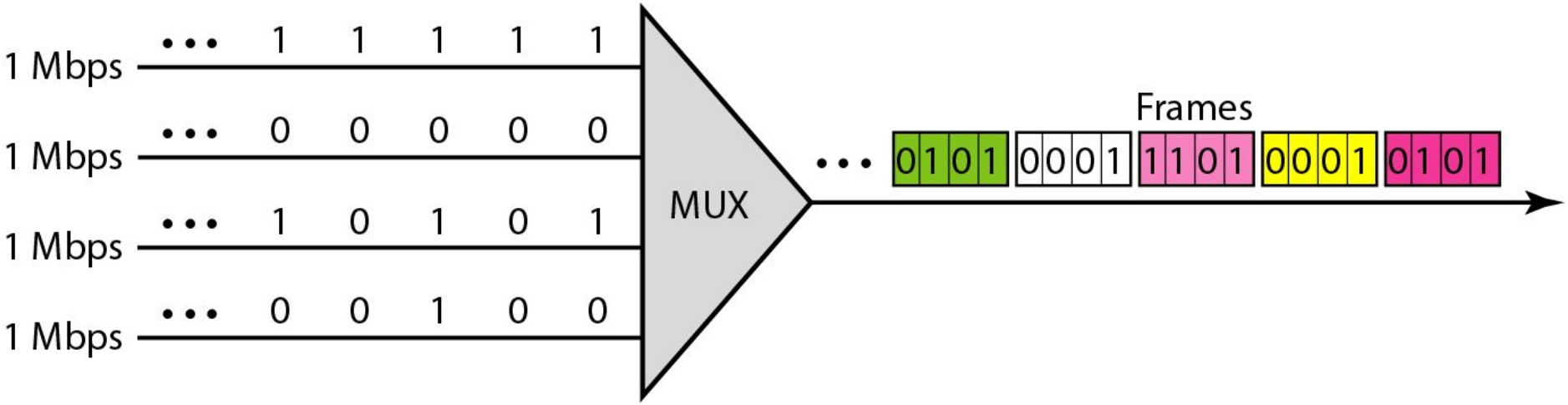
b. The output bit duration is one-fourth of the input bit duration, or ¼ µs.

Example 6.6 (continued)

c. ***The output bit rate is the inverse of the output bit duration or 1/(4μs) or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.***

d. ***The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.***

Figure 6.14 *Example 6.6*



Example 6.7

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame.

Solution

We can answer the questions as follows:

a. The duration of 1 bit before multiplexing is 1 / 1 kbps, or 0.001 s (1 ms).

b. The rate of the link is 4 times the rate of a connection, or 4 kbps.

Example 6.7 (continued)

c. ***The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or 1/4 ms or 250 μs. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or 1/4 kbps or 250 μs.***

d. ***The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μs, or 1 ms.***

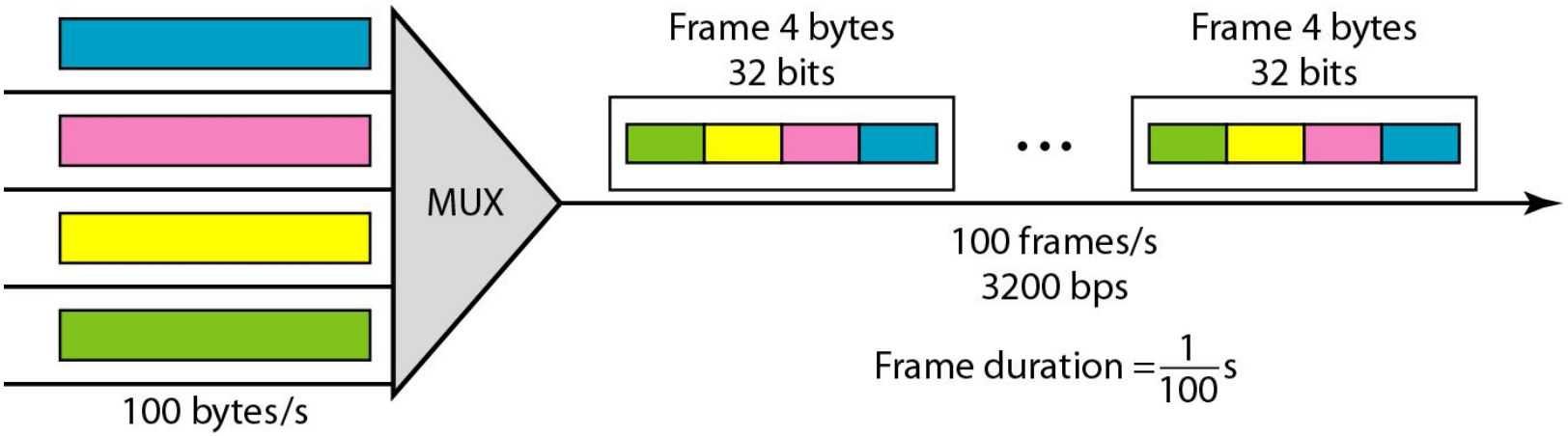
Example 6.8

Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Solution

The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second. The bit rate is 100 × 32, or 3200 bps.

Figure 6.16 *Example 6.8*



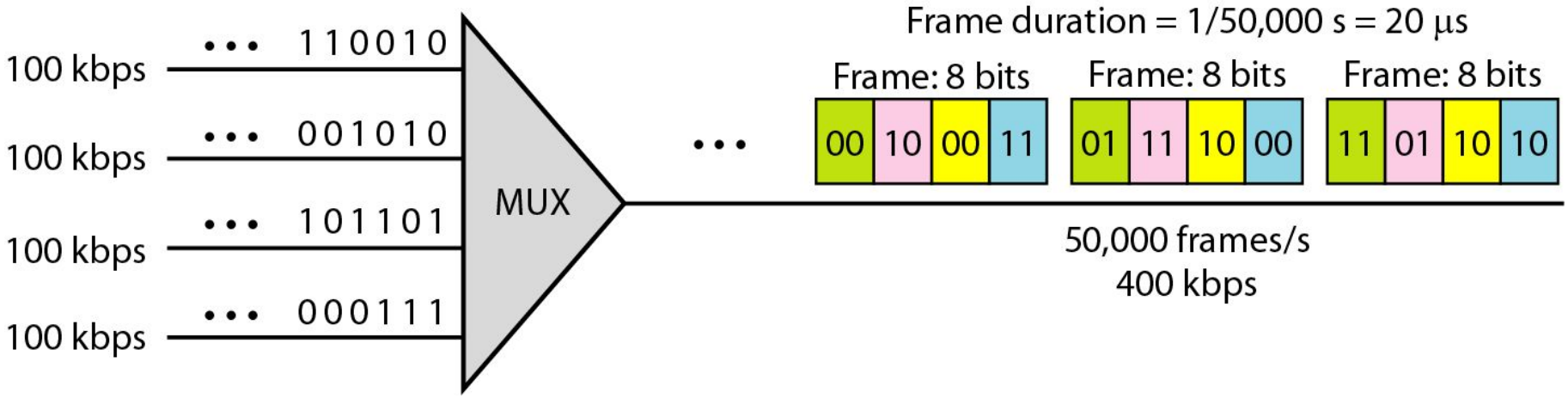
Example 6.9

A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

Solution

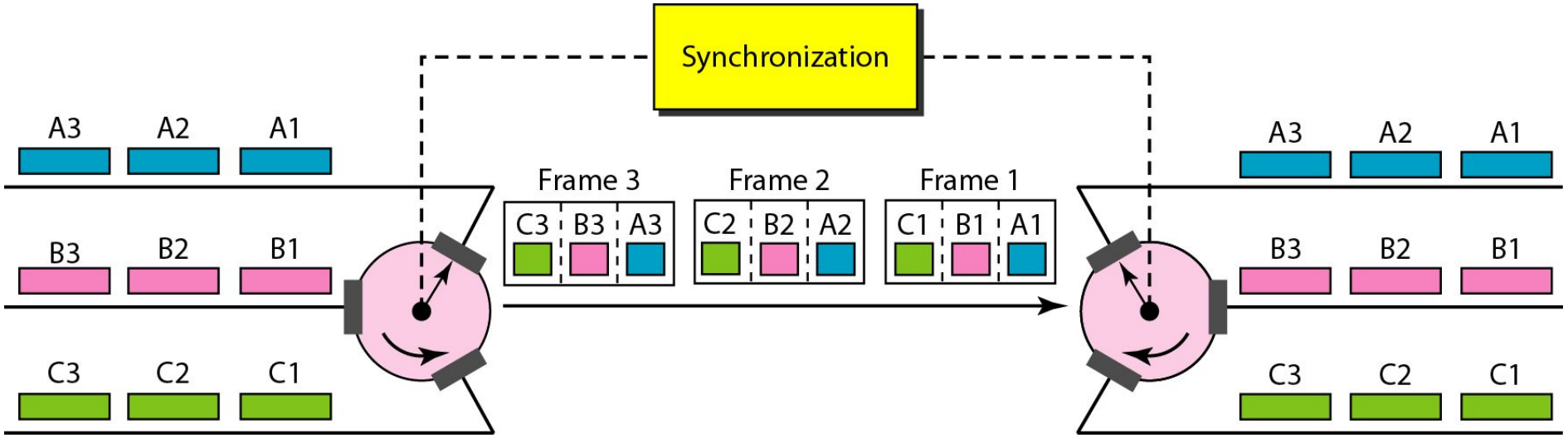
Figure 6.17 shows the output (4x100kbps) for four arbitrary inputs. The link carries 400K/(2x4)=50,000 2x4=8bit frames per second. The frame duration is therefore 1/50,000 s or 20 µs. The bit duration on the output link is 1/400,000 s, or 2.5 µs.

Figure 6.17 *Example 6.9*



Interleaving

- The process of taking a group of bits from each input line for multiplexing is called interleaving.
- TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side.
- The switches are synchronized and rotate at the same speed, but in opposite directions.
- On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path. This process is called interleaving.

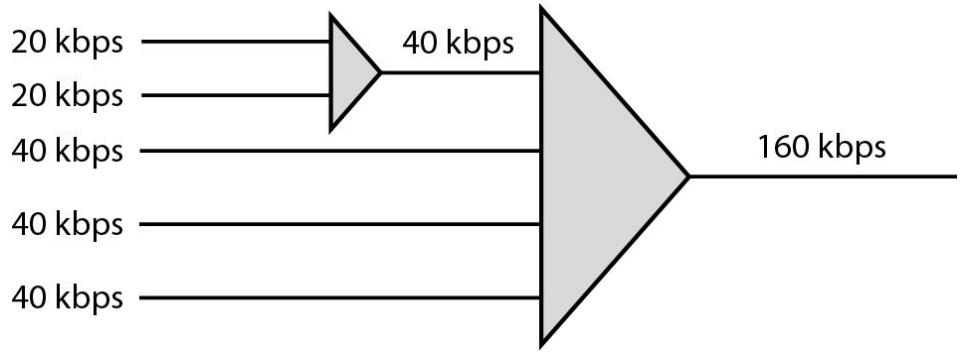


Data Rate Management

- Not all input links maybe have the same data rate.
- Some links maybe slower. There maybe several different input link speeds
- There are three strategies that can be used to overcome the data rate mismatch: multilevel, multislot and pulse stuffing

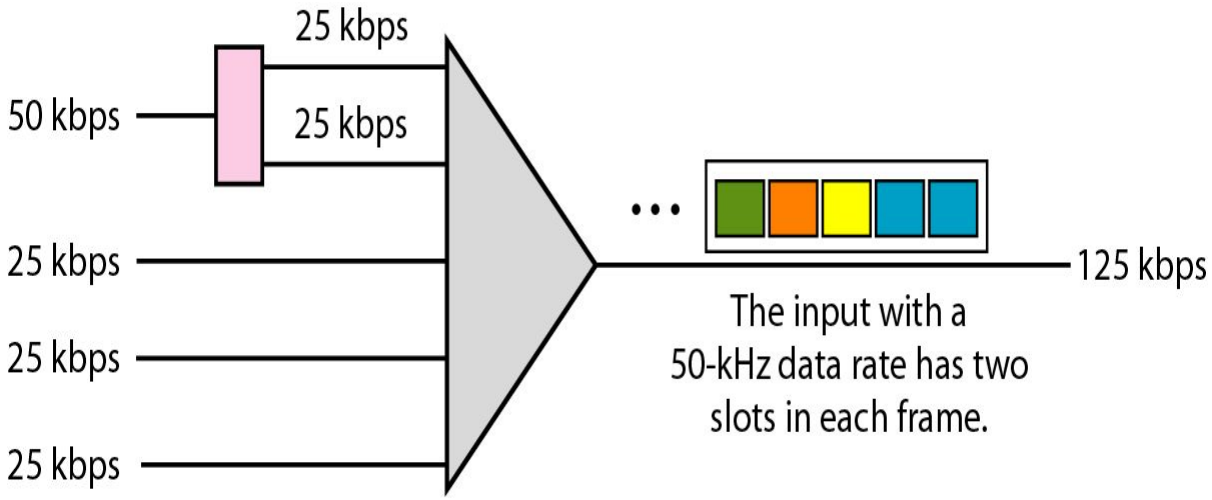
Multilevel multiplexing

Multilevel multiplexing *is a technique used when the data rate of an input line is a multiple of others. For example, in Figure, we have two inputs of 20 kbps and three inputs of 40 kbps. The first two input lines can be multiplexed together to provide a data rate equal to the last three. A second level of multiplexing can create an output of 160 kbps.*



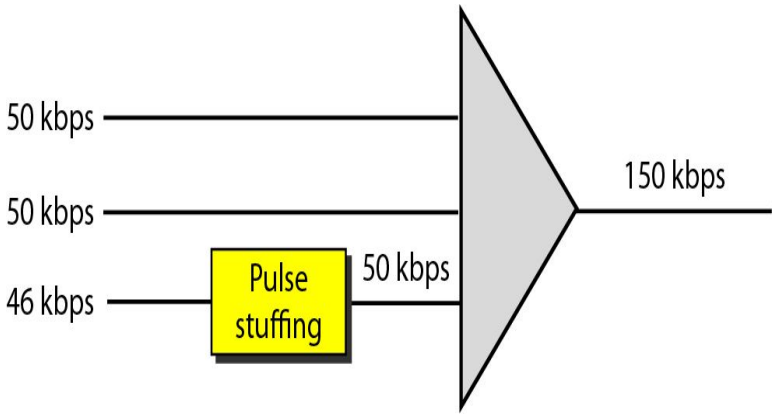
Multiple-Slot Allocation

Sometimes it is more efficient to allot more than one slot in a frame to a single input line. For example, we might have an input line that has a data rate that is a multiple of another input. In Figure, the input line with a 50-kbps data rate can be given two slots in the output. We insert a serial-to-parallel converter in the line to make two inputs out of one.



Pulse Stuffing

Sometimes the bit rates of sources are not multiple integers of each other. Therefore, neither of the above two techniques can be applied. One solution is to make the highest input data rate the dominant data rate and then add dummy bits to the input lines with lower rates. This will increase their rates. This technique is called pulse stuffing, bit padding, or bit stuffing. The input with a data rate of 46 is pulse-stuffed to increase the rate to 50 kbps. Now multiplexing can take place.



Example 6.11

Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?

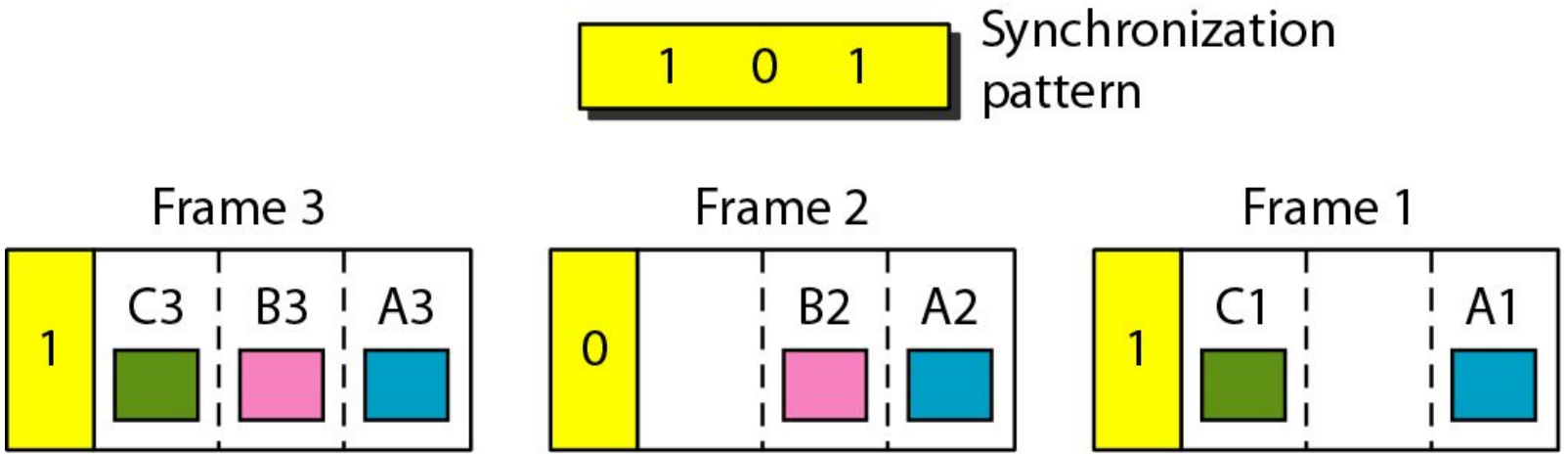
Solution

We can allocate one slot to the first channel and two slots to the second channel. Each frame carries 3 bits. The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. The bit rate is 100,000 frames/s × 3 bits per frame, or 300 kbps.

Frame Synchronization

- To ensure that the receiver correctly reads the incoming bits, i.e., knows the incoming bit boundaries to interpret a “1” and a “0”, a known bit pattern is used between the frames.
- The receiver looks for the anticipated bit and starts counting bits till the end of the frame.
- Then it starts over again with the reception of another known bit.
- These bits (or bit patterns) are called synchronization bit(s).
- They are part of the overhead of transmission.

Figure 6.22 *Framing bits*



Example 6.10

We have four sources, each creating 250 8-bit characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.

Solution

We can answer the questions as follows:

a. The data rate of each source is 250 × 8 = 2000 bps = 2 kbps.

Example 6.10 (continued)

b. Each source sends 250 characters per second; therefore, the duration of a character is 1/250 s, or 4 ms.

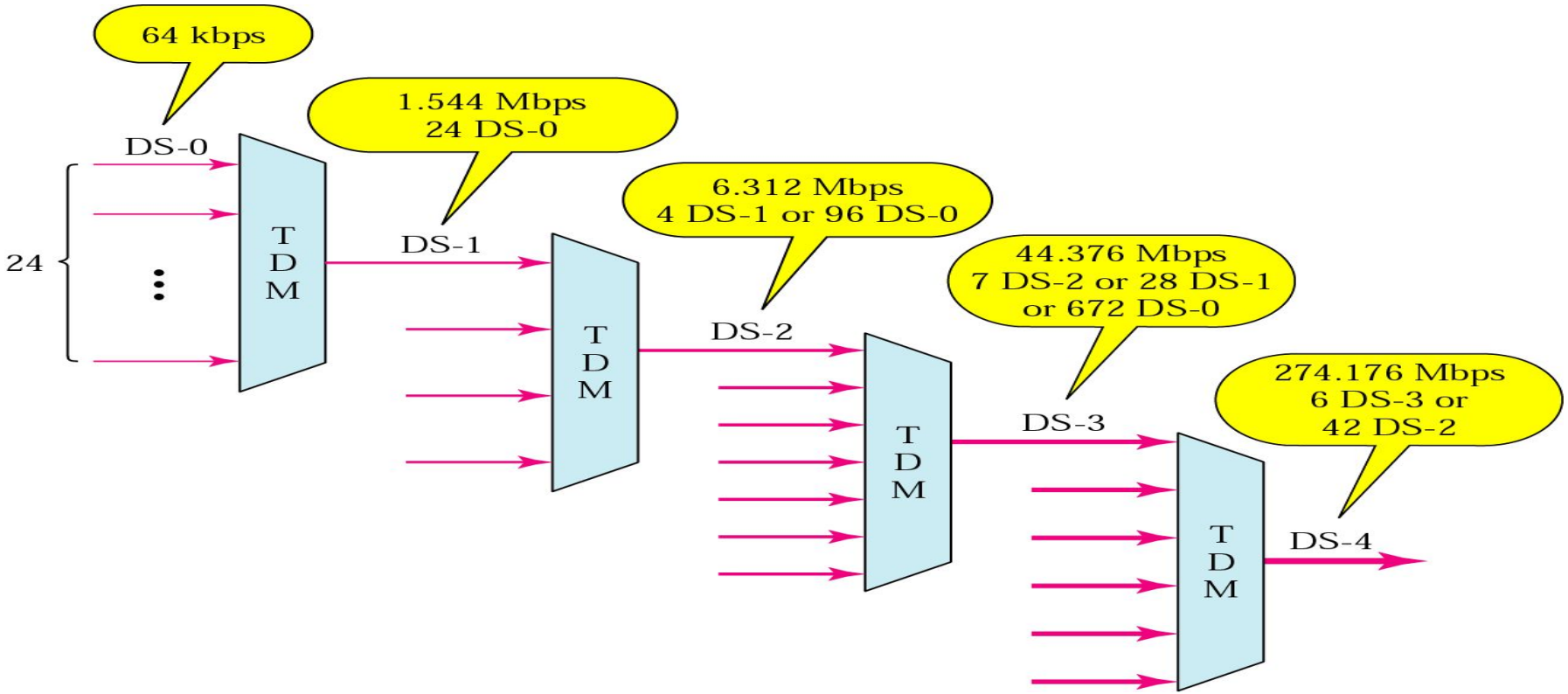
c. Each frame has one character from each source, which means the link needs to send 250 frames per second to keep the transmission rate of each source.

d. The duration of each frame is 1/250 s, or 4 ms. Note that the duration of each frame is the same as the duration of each character coming from each source.

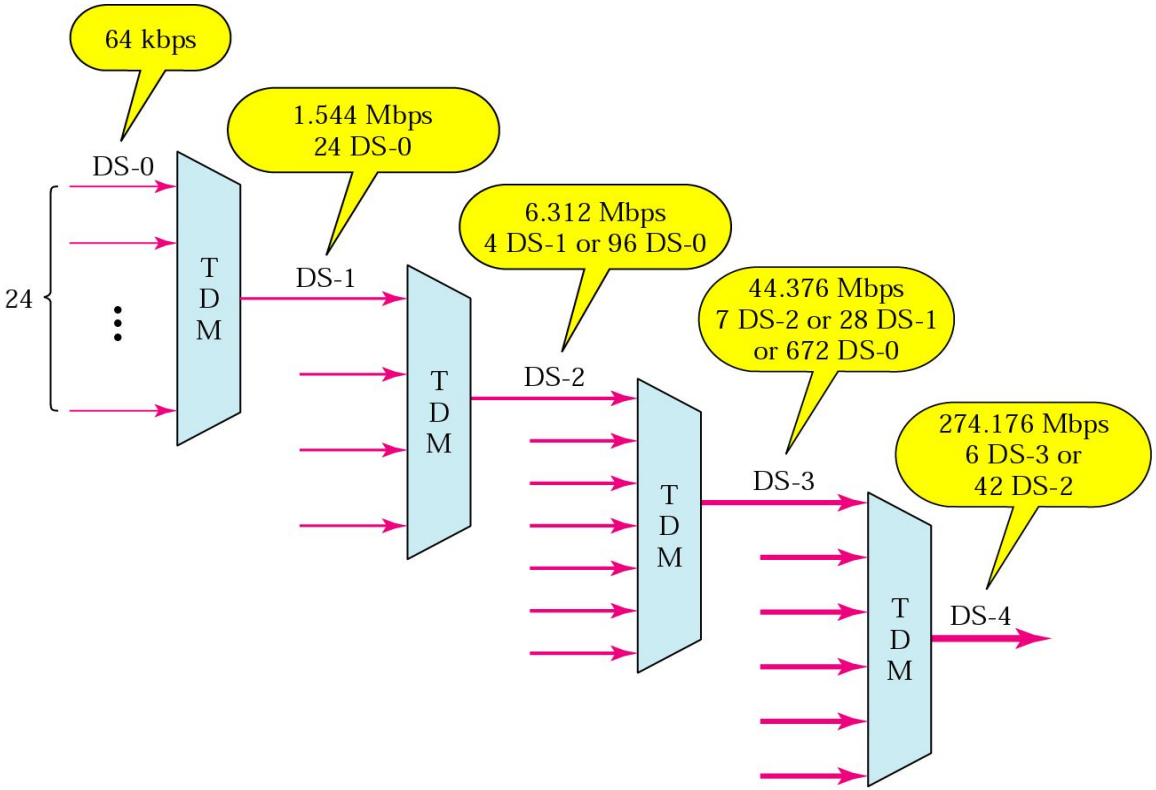
e. Each frame carries 4 characters and 1 extra synchronizing bit. This means that each frame is $4 \times 8 + 1 = 33$ bits.

DS hierarchy

Telephone companies implement TDM through a hierarchy of digital signals, called digital signal (DS) service or digital hierarchy. Figure shows the data rates supported by each level.

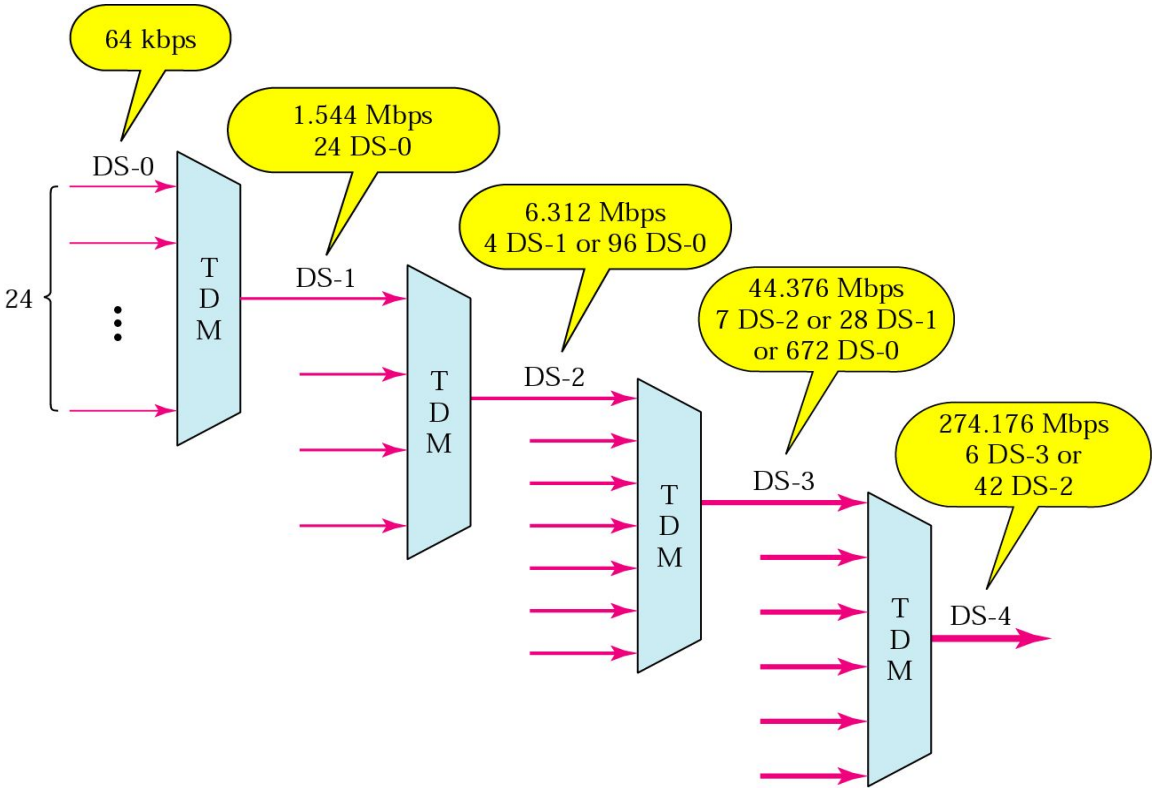


Cont.



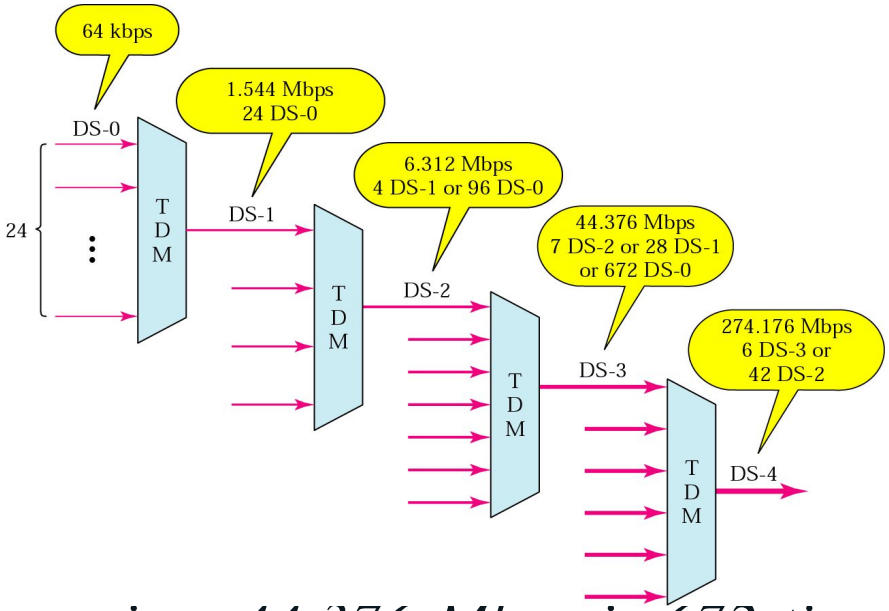
A DS-0 service is a single digital channel of 64 kbps. DS-1 is a 1.544-Mbps service; 1.544 Mbps is 24 times 64 kbps plus 8 kbps of overhead. It can be used as a single service for 1.544-Mbps transmissions, or it can be used to multiplex 24 DS-0 channels or to carry any other combination desired by the user that can fit within its 1.544-Mbps capacity.

Cont.



DS-2 is a 6.312-Mbps service; 6.312 Mbps is 96 times 64 kbps plus 168 kbps of overhead. It can be used as a single service for 6.312-Mbps transmissions; or it can be used to multiplex 4 DS-l channels, 96 DS-0 channels, or a combination of these service types.

Cont.



DS-3 is a 44.376-Mbps service; 44.376 Mbps is 672 times 64 kbps plus 1.368 Mbps of overhead. It can be used as a single service for 44.376-Mbps transmissions; or it can be used to multiplex 7 DS-2 channels, 28 DS-l channels, 672 DS-0 channels, or a combination of these service types.

DS-4 is a 274.176-Mbps service; 274.176 is 4032 times 64 kbps plus 16.128 Mbps of overhead. It can be used to multiplex 6 DS-3 channels, 42 DS-2 channels, 168 DS-l channels, 4032 DS-0 channels, or a combination of these service types

Table: DS and T lines rates

- *DS-0, DS-l, and so on are the names of services. To implement those services, the telephone companies use **T lines** (T-l to T-4).*
- *These are lines with capacities precisely matched to the data rates of the DS-l to DS-4 services.*
- *So far only T-l and T-3 lines are commercially available.*
- *The T-l line is used to implement DS-l; T-2 is used to implement DS-2; and so on.*
- *DS-0 is not actually offered as a service, but it has been defined as a basis for reference purposes.*

Service	Line	Rate (Mbps)	Voice Channels
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

E1 line

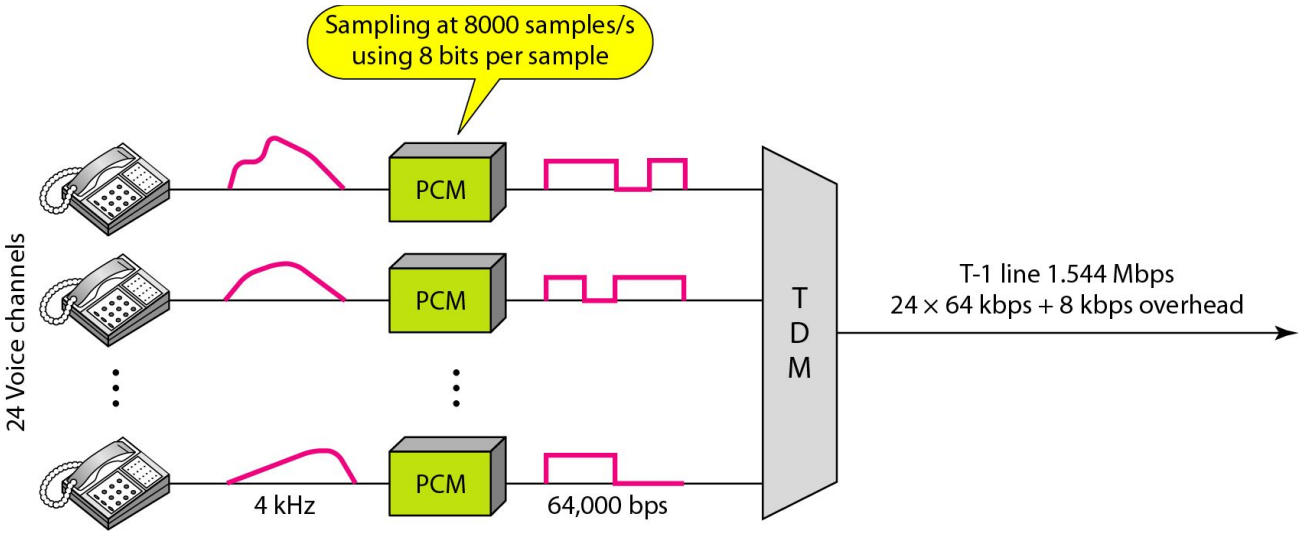
*Europeans use a version of **T** lines called **E** lines. The two systems are conceptually identical, but their capacities differ. Table below shows the **E** lines and their capacities.*

Line	*Rate (Mbps)*	*Voice Channels*
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

T Lines for Analog Transmission

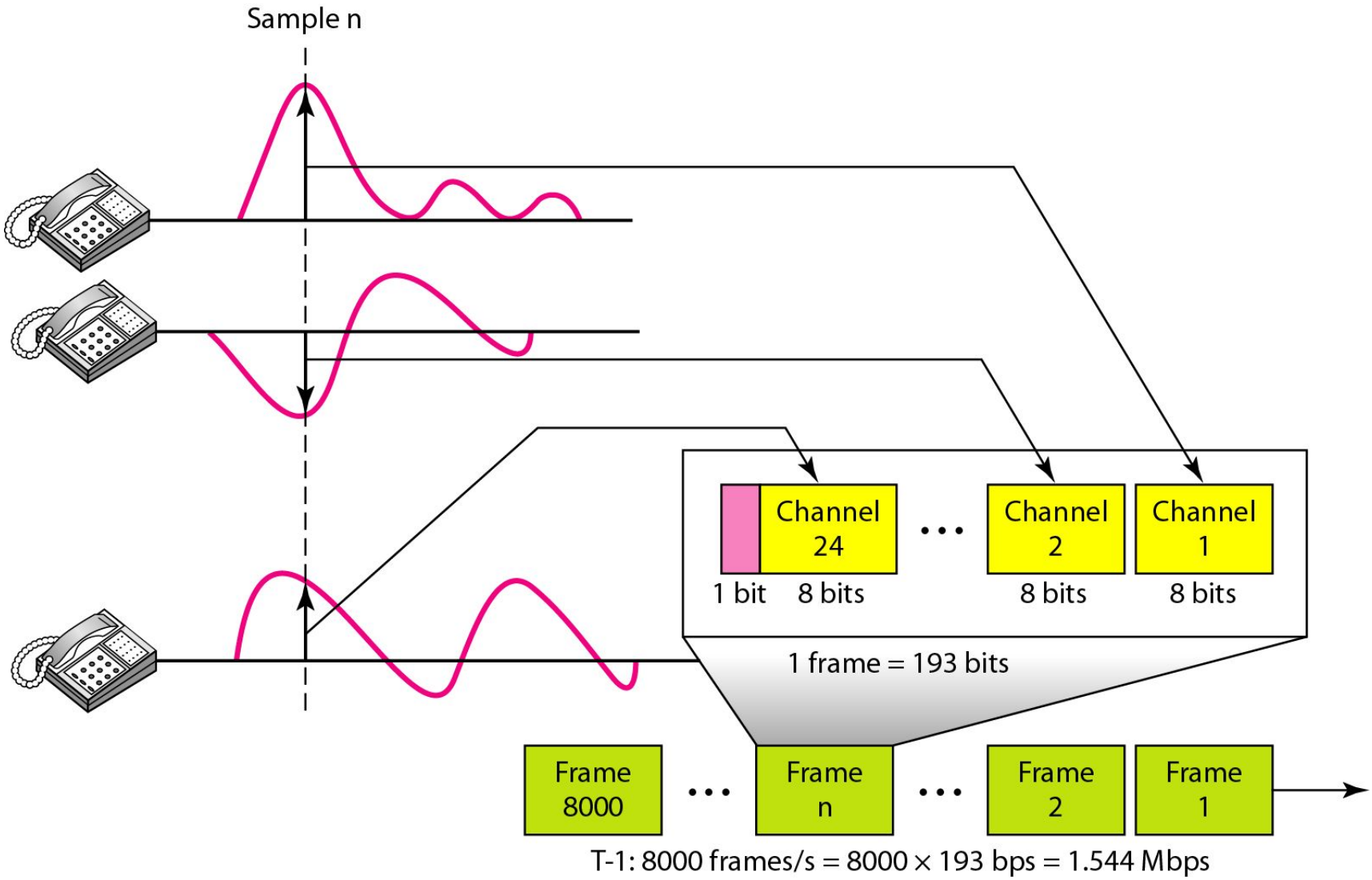
- *T lines are digital lines designed for the transmission of digital data, audio, or video.*
- *However, they also can be used for analog transmission (regular telephone connections), provided the analog signals are first sampled, then time-division multiplexed.*
- *The possibility of using T lines as analog carriers opened up a new generation of services for the telephone companies.*
- *Earlier, when an organization wanted 24 separate telephone lines, it needed to run 24 twisted-pair cables from the company to the central exchange.*
- *Today, that same organization can combine the 24 lines into one T-l line and run only the T-l line to the exchange.*

T Lines for Analog Transmission-Cont.



- *The frame used on a T-l line is usually 193 bits divided into 24 slots of 8 bits each plus 1 extra bit for synchronization (24 × 8+ 1 = 193).*
- *In other words, each slot contains one signal segment from each channel; 24 segments are interleaved in one frame.*
- *If a T-l line carries 8000 frames/sec(One frame for each sampling period)*
- *The data rate is 1.544 Mbps (193 × 8000 = 1.544 Mbps)-the capacity of the line.*

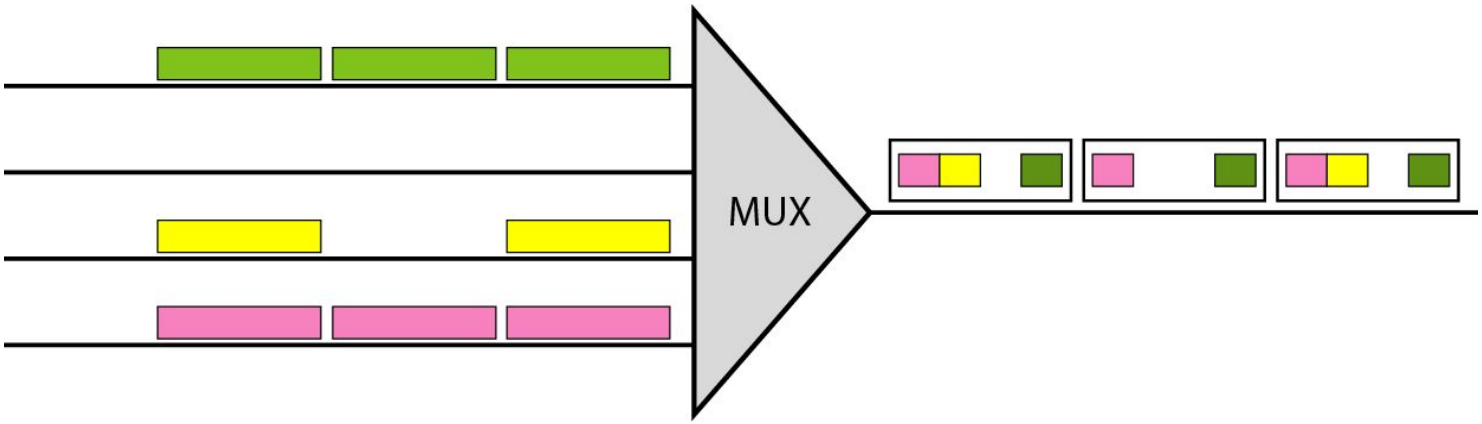
T-1 frame structure



Statistical time division multiplexing

Inefficient use of Bandwidth in Synchronous Time Division Multiplexing

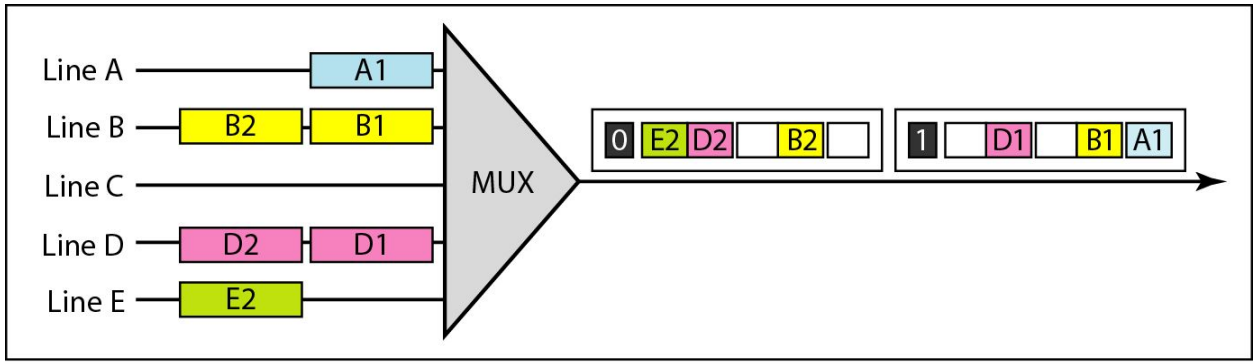
- *Sometimes an input link may have no data to transmit.*
- *When that happens, one or more slots on the output link will go unused.*
- *That is wasteful of bandwidth*



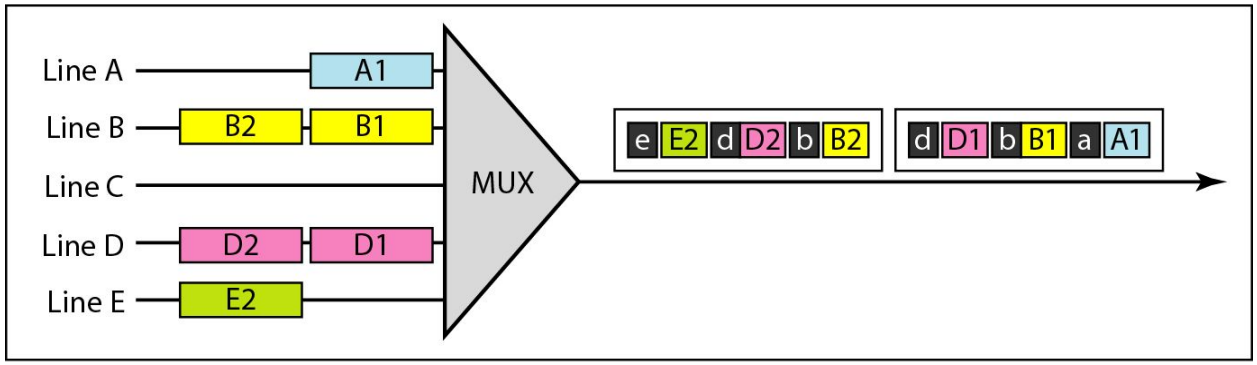
TDM slot comparison

In synchronous TDM, each input has a reserved slot in the output frame.
This can be inefficient if some input lines have no data to send.
In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.
Only when an input line has a slot's worth of data to send is it given a slot in the output frame.
In statistical multiplexing, the number of slots in each frame is less than the number of input lines. The multiplexer checks each input line in round robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.

Cont.



a. Synchronous TDM



b. Statistical TDM

Addressing

- *Above figure also shows a major difference between slots in synchronous TDM and statistical TDM.*
- *An output slot in synchronous TDM is totally occupied by data; in statistical TDM, a slot needs to carry data as well as the address of the destination.*
- *In synchronous TDM, there is no need for addressing; synchronization and pre-assigned relationships between the inputs and outputs serve as an address.*
- *In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no pre-assigned or reserved slots.*
- *We need to include the address of the receiver inside each slot to show where it is to be delivered. The addressing in its simplest form can be n bits to define N different output lines with $n=\log_2 N$. For example, for eight different output lines, we need a 3-bit address.*

Slot Size

Since a slot carries both data and an address in statistical TDM, the ratio of the data size to address size must be reasonable to make transmission efficient.

For example, it would be inefficient to send 1 bit per slot as data when the address is 3 bits.

This would mean an overhead of 300 percent.

In statistical TDM, a block of data is usually many bytes while the address is just a few bytes.

No Synchronization Bit

There is another difference between synchronous and statistical TDM, but this time it is at the frame level. The frames in statistical TDM need not be synchronized, so we do not need synchronization bits.

Bandwidth

In statistical TDM, the capacity of the link is normally less than the sum of the capacities of each channel. The designers of statistical TDM define the capacity of the link based on the statistics of the load for each channel. If on average only x percent of the input slots are filled, the capacity of the link reflects this. Of course, during peak times, some slots need to wait.